

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION
CLASS IX • SCIENCE (SUBJECT CODE: 086) • ACADEMIC SESSION 2026-27

SET D — ASSERTION & REASONING SPECIAL

CHAPTER 03: TISSUES: PLANT & ANIMAL STRUCTURES

TIME ALLOWED: 45 MINUTES MAXIMUM MARKS: 25 MARKS SERIES: CBS-IX-SCI-SET-D-03

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

GENERAL INSTRUCTIONS (SET D - ASSERTION & REASONING SPECIAL):

1. The question paper comprises **25 Assertion-Reasoning Questions** of **1 mark each**.
2. Each question consists of two statements — **Assertion (A)** and **Reason (R)**.
3. Select the correct option from the following 4 standardized choices:
 - **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
 - **(B)** Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
 - **(C)** Assertion (A) is true but Reason (R) is false.
 - **(D)** Assertion (A) is false but Reason (R) is true.
4. **Section A (Q1 – Q10):** Core Theoretical Principles & Scientific Definitions.
5. **Section B (Q11 – Q18):** Experimental Observations & Phenomenological Inferences.
6. **Section C (Q19 – Q25):** Advanced Conceptual Nuances & Common Scientific Misconceptions.
7. All questions are compulsory. There is no negative marking.
8. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

STANDARD DIRECTIONS FOR ALL 25 ASSERTION & REASONING QUESTIONS:

- (A)** Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

SECTION A: CORE THEORETICAL PRINCIPLES (Q.1 TO Q.10)

[10 MARKS]

Q.1

Assertion (A): Meristematic tissues in plants are characterized by continuously dividing, undifferentiated cells.

Reason (R): Meristematic cells possess dense cytoplasm, thin cellulose walls, prominent nuclei, and lack vacuoles.

- (A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.2

Assertion (A): Apical meristem is responsible for the increase in length of plant stems and roots.

Reason (R): Apical meristem is located at the growing tips of stems and roots and adds new primary cells longitudinally.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.3

Assertion (A): The girth (thickness) of the stem or root of a woody dicot plant increases due to lateral meristem (cambium).

Reason (R): Lateral meristems undergo periclinal cell divisions, adding secondary xylem and phloem radially to the vascular cylinder.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.4

Assertion (A): Sclerenchyma cells provide immense mechanical strength and rigidity to plant organs.

Reason (R): Sclerenchyma tissues are composed of dead cells with uniformly thick walls heavily impregnated with lignin.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.5

Assertion (A): Collenchyma provides both mechanical support and flexibility to young dicot stems and leaf petioles.

Reason (R): Collenchyma cells have localized wall thickenings of cellulose, hemicellulose, and pectin at their cell corners.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.6

Assertion (A): Xylem is classified as a complex permanent tissue in vascular plants.

Reason (R): Xylem consists of multiple distinct types of cells (tracheids, vessels, xylem parenchyma, and xylem fibres) working together as a functional unit.

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| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

Q.7

Assertion (A): Blood is classified as a connective tissue.

Reason (R): Blood has a fluid matrix called plasma in which RBCs, WBCs, and platelets are suspended, connecting organ systems via transport.

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| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

Q.8

Assertion (A): Cardiac muscle tissue can contract rhythmically throughout a person's life without ever experiencing muscular fatigue.

Reason (R): Cardiac muscle fibers are involuntary, cylindrical, branched, uninucleate, and richly supplied with mitochondria and capillaries.

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|--|--|
| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

Q.9

Assertion (A): Skeletal muscles are also termed voluntary striated muscles.

Reason (R): Their contractions are under conscious voluntary nervous control and display alternating dark and light bands under a light microscope.

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| (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). | (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A). |
| (C) Assertion (A) is true but Reason (R) is false. | (D) Assertion (A) is false but Reason (R) is true. |

Q.10

Assertion (A): Tendons connect bone to bone across a movable joint.

Reason (R): Tendons are tough, fibrous connective tissues with high tensile strength and limited elasticity.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

SECTION B: EXPERIMENTAL OBSERVATIONS & INFERENCES (Q.11 TO Q.18)

[8 MARKS]

Q.11

Assertion (A): Aquatic plants like *Hydrilla* and water hyacinth float effortlessly on the surface of water.

Reason (R): Parenchyma in aquatic plants modifies into aerenchyma containing large interconnected air cavities that provide buoyancy.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.12

Assertion (A): The husk of a dry coconut is extremely tough, stiff, and fibrous.

Reason (R): Coconut husk is made of dense sclerenchymatous fibres with heavily lignified, cemented cell walls.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.13

Assertion (A): Stomata in the leaf epidermis open and close during gas exchange and transpiration.

Reason (R): Each stomatal pore is bounded by two kidney-shaped guard cells whose turgor changes cause stomatal opening and closing.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.14

Assertion (A): Desert plants (xerophytes) have a thick waxy cuticle layer of cutin covering their outer epidermis.

Reason (R): The hydrophobic waxy cutin layer drastically minimizes excessive water loss through transpiration in arid environments.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.15

Assertion (A): A fractured bone can repair itself over time, whereas severe cartilage wear at joint surfaces heals very poorly.

Reason (R): Bone is richly vascularized with living osteocytes and blood capillaries in Haversian canals, whereas cartilage is largely avascular.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.16

Assertion (A): Ciliated columnar epithelium lines the human respiratory tract and fallopian tubes.

Reason (R): The rhythmic, coordinated beating of cilia moves mucus and trapped foreign particles outwards along the tract surface.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.17

Assertion (A): Mature sieve tube elements in phloem can conduct organic nutrients even though they lack a cell nucleus.

Reason (R): Each sieve tube element is intimately connected to an adjacent nucleated companion cell that regulates its metabolic activities.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.18

Assertion (A): Cork acts as an impervious protective outer covering on mature tree trunks.

Reason (R): The cell walls of mature dead cork tissue contain suberin, a chemical that makes them completely impermeable to water and gases.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

SECTION C: ADVANCED NUANCES & MISCONCEPTIONS (Q.19 TO Q.25)

[7 MARKS]

Q.19

Assertion (A): Smooth (unstriated) muscles are present in the walls of the stomach, intestine, and blood vessels.

Reason (R): The contractions of visceral internal organs are involuntary and do not show cross-striations under a microscope.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.20

Assertion (A): Ligaments are strong, flexible connective tissues that hold bones together at joints.

Reason (R): Ligaments contain a high proportion of elastin fibres, imparting considerable elasticity while preventing joint dislocation.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.21

Assertion (A): Xylem parenchyma is the only living component in the xylem tissue of angiosperms.

Reason (R): Tracheids, vessels, and xylem fibres are all dead, lignified cells at functional maturity.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false. (D) Assertion (A) is false but Reason (R) is true.

Q.22

Assertion (A): Nerve impulses travel rapidly over long distances across the human body.

Reason (R): Neurons possess a long transmitting cytoplasmic extension called the axon, often covered with an insulating myelin sheath.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.23

Assertion (A): Adipose tissue acts as an efficient thermal insulator beneath the skin.

Reason (R): Adipocytes store fat globules, and subcutaneous fat is a poor conductor of heat, preventing heat loss from the core body.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.24

Assertion (A): Intercalary meristem present at the base of leaves or internodes allows grasses to regenerate quickly after grazing by herbivores.

Reason (R): Intercalary meristem cells divide actively, regenerating lost shoot tissue from below the clipped tip.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.25

Assertion (A): Epithelial tissue forms a continuous protective sheet with no intercellular spaces or blood vessels.

Reason (R): Epithelial cells rest on an extracellular basement membrane that separates them from underlying connective tissue.

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET D)

01	☐ A ☐ B ☐ C ☐ D	02	☐ A ☐ B ☐ C ☐ D	03	☐ A ☐ B ☐ C ☐ D	04	☐ A ☐ B ☐ C ☐ D	05	☐ A ☐ B ☐ C ☐ D
06	☐ A ☐ B ☐ C ☐ D	07	☐ A ☐ B ☐ C ☐ D	08	☐ A ☐ B ☐ C ☐ D	09	☐ A ☐ B ☐ C ☐ D	10	☐ A ☐ B ☐ C ☐ D
11	☐ A ☐ B ☐ C ☐ D	12	☐ A ☐ B ☐ C ☐ D	13	☐ A ☐ B ☐ C ☐ D	14	☐ A ☐ B ☐ C ☐ D	15	☐ A ☐ B ☐ C ☐ D
16	☐ A ☐ B ☐ C ☐ D	17	☐ A ☐ B ☐ C ☐ D	18	☐ A ☐ B ☐ C ☐ D	19	☐ A ☐ B ☐ C ☐ D	20	☐ A ☐ B ☐ C ☐ D
21	☐ A ☐ B ☐ C ☐ D	22	☐ A ☐ B ☐ C ☐ D	23	☐ A ☐ B ☐ C ☐ D	24	☐ A ☐ B ☐ C ☐ D	25	☐ A ☐ B ☐ C ☐ D

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & SCIENTIFIC CAUSAL EXPLANATIONS (SET D)

Q. No.	Correct	Scientific Rationale & Causal Analysis
Q.1	(B)	Both statements are true, but Reason describes the cytological features rather than explaining why they continuously divide.
Q.2	(A)	Assertion is true. Reason is true and explains the primary growth function of the shoot and root apical meristems.
Q.3	(A)	Assertion is true. Reason is true and explains the mechanism of secondary thickening.
Q.4	(A)	Assertion is true. Reason is true and explains how lignified, thick dead walls provide mechanical support (e.g. coconut husk).
Q.5	(A)	Assertion is true. Reason is true and explains the structural basis of bending without breaking.
Q.6	(A)	Assertion is true. Reason is true and defines why it is classified as a 'complex' tissue rather than simple tissue.
Q.7	(A)	Assertion is true. Reason is true and explains why blood fulfills the histological definition of connective tissue.
Q.8	(A)	Assertion is true. Reason is true and explains how continuous energy supply and branched syncytial structure prevent fatigue.
Q.9	(A)	Assertion is true. Reason is true and explains both descriptive terms: 'voluntary' and 'striated'.
Q.10	(D)	Assertion is false: Ligaments connect bone to bone; tendons connect muscle to bone. Reason is true (tendons are tough fibrous cords with limited elasticity).
Q.11	(A)	Assertion is true. Reason is true and explains how aerenchyma air pockets reduce density and impart buoyancy.
Q.12	(A)	Assertion is true. Reason is true and provides the structural tissue responsible for the fibrous rigidity of coconut husk.
Q.13	(A)	Assertion is true. Reason is true and explains the biomechanical action of guard cells regulating stomatal aperture.
Q.14	(A)	Assertion is true. Reason is true and explains the adaptive physiological function of cutin.
Q.15	(A)	Assertion is true. Reason is true and explains why bone has high regenerative capacity while avascular cartilage heals slowly.
Q.16	(A)	Assertion is true. Reason is true and directly explains the directional transport function of ciliated epithelial cells.
Q.17	(A)	Assertion is true. Reason is true and explains the vital symbiotic dependence of enucleated sieve tubes on companion cells.
Q.18	(A)	Assertion is true. Reason is true and explains how suberin deposition creates a protective, waterproof barrier.
Q.19	(A)	Assertion is true. Reason is true and explains why these visceral structures contain unstriated, involuntary smooth muscle fibers.
Q.20	(A)	Assertion is true. Reason is true and describes how elastin provides flexibility and joint stabilization.
Q.21	(A)	Assertion is true. Reason is true and confirms that all other three xylem elements are dead at maturity.
Q.22	(A)	Assertion is true. Reason is true and explains how axonal architecture enables rapid electrical conduction.
Q.23	(A)	Assertion is true. Reason is true and explains the physiological thermal insulation mechanism of adipose tissue.
Q.24	(A)	Assertion is true. Reason is true and explains the evolutionary advantage of intercalary meristem in grass survival.
Q.25	(B)	Both statements are true facts of epithelial histology, but Reason describes the basement membrane attachment rather than explaining why cells are packed tightly without intercellular gaps.